HOW AZURE'S MODEL ROUTER BOOSTS LLM EFFICIENCY

UNLOCKING SMARTER AI WORKLOADS WITH AZURE FOUNDRY

CENTRALIZE MODEL ACCESS, DECENTRALIZE MODEL CHOICE

ABOUT US

- Cloud & DevOps Consultant with 15+ years of Industry Experience
- Working to build a Next Gen AI Platform
- Active on MLOps, AIOps space

- Lead DevOps Architect with 10+ years of experience in cloud and platform engineering
- I work in DevOps and cloud architecture, focusing on Kubernetes, automation, and scalable systems.

WHAT'S INSIDE

WHAT WE'LL COVER IN THE NEXT 15 MINUTES

- The Problem
- How this Helps!
- Demo Time
- Lessons Learnt

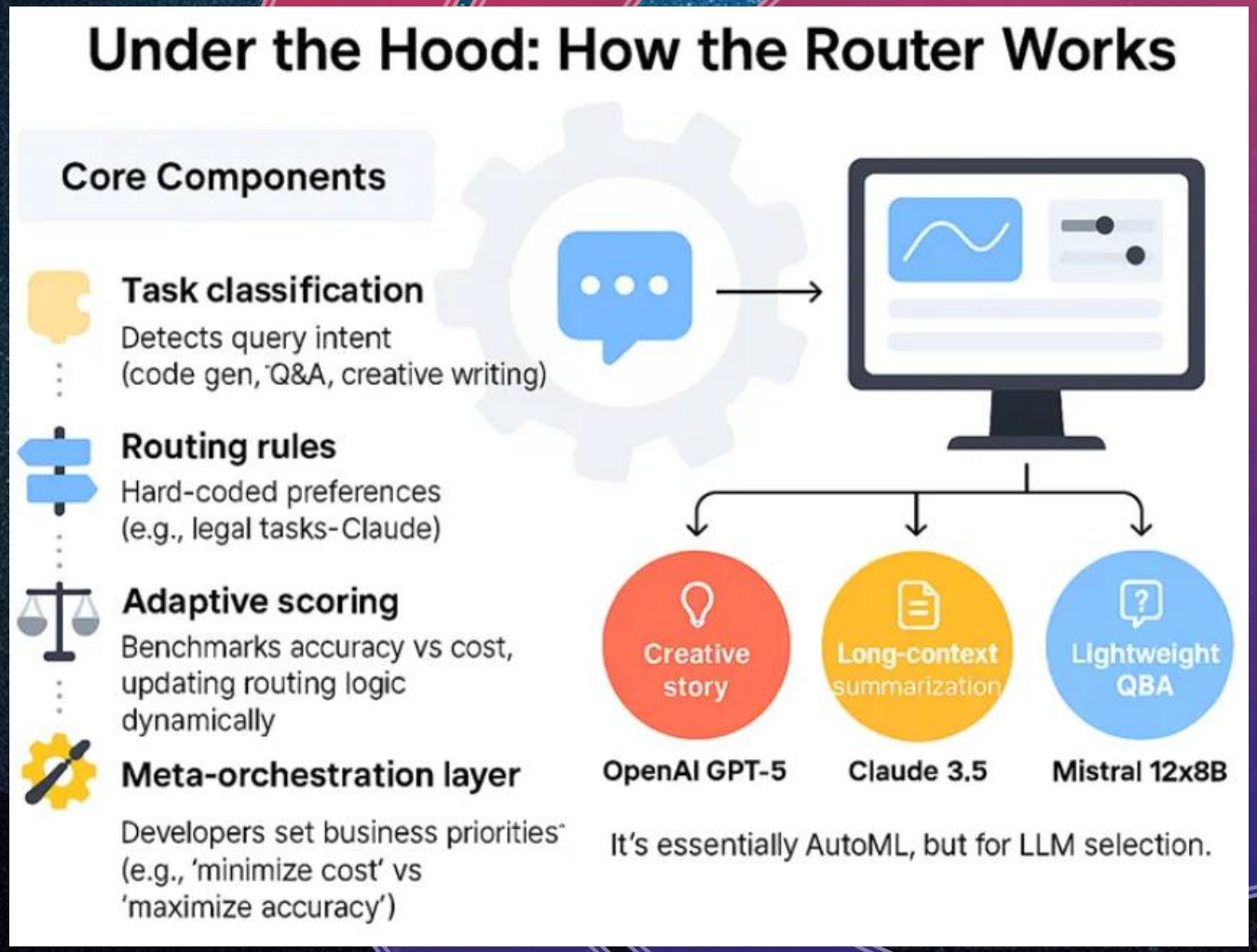
THE PROBLEM

- One model does NOT fit ALL
  - Wasted Reasoning Capacity
  - Quality drop on complexity

- Maintainability Issue
  - Model expiry
  - Manual routing = Maintenance Nightmare

- Cost Benefits
  - Latency Reduction
  - Less overpaying

HOW THIS HELPS US!

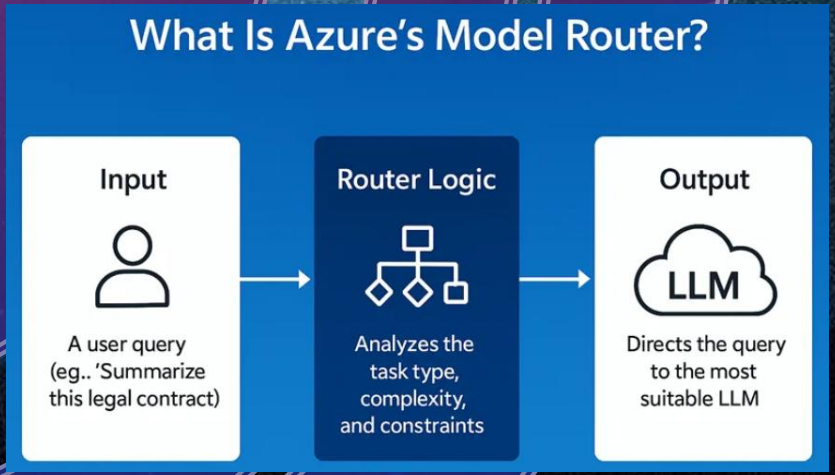


100% Transparency

18+ Models, 1 Deployment

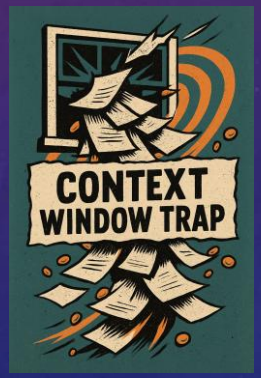
3 Routing Modes
- Cost
- Balanced
- Quality

1 API,
0 Code Changes



LESSONS LEARNT

- Quick v/s Custom Deployments
  - Experiment & finalize – Swap new LLM's
  - Build on Automation – ***Terraform, Bicep Templates***
- Don't Fly Blind
  - Track Latency, Cost & Usage.
  - Global v/s Data Zone
- Use a Hybrid Approach
  - Model Router for generic use-cases v/s targeted models for specific ones.
  - ***Be Cautious with Claude***
- Context Window Trap
  - Router might select Nano for large docs.
  - Intermittent Failures



DEMO

GITHUB

THANK YOU!

SOUMADEEP BHATTACHARYA

ANIK GUHA